

PROJECT REPORT

Emotion Engine

Automated Facial Emotion Recognition System

Prepared by: Raj Pratap Singh Sengar

Date: 6 August 2026

Status: Core system complete — pilot-ready

1. Executive Summary

This project developed a system that automatically reads a person's emotional expression — happy, sad, angry, surprised, and others — directly from a photo or a live camera feed, without any human needing to manually review the image. We trained and compared six different AI approaches, selected the strongest performer, and built it into a working application with two ways to use it: uploading a photo, or a live webcam view.

The final system correctly identifies the right emotion roughly two times out of three (63%) — in line with published benchmarks for this task, and close to the natural ceiling for how consistently even humans agree on labelling ambiguous expressions in the same standard dataset. It performs strongly on clearer, more common emotions such as happiness and surprise, and reasonably on more subtle ones.

The core model and both application interfaces (photo upload and live webcam) are complete and working today. This gives us a ready foundation to pilot in a real use case — such as gauging customer sentiment in service interactions, supporting wellbeing check-ins, or making digital interfaces more responsive to how a user is feeling — before investing further in scaling or specialization.

Key Takeaways

- **A working, end-to-end emotion recognition system was delivered — model, live webcam app, and web interface — not just a research prototype.**
- **Accuracy (63%) is in line with independent published benchmarks for this exact task, not a below-standard result.**
- **The system is ready for a low-risk pilot; broader deployment would benefit from additional real-condition data, covered in Section 8.**

2. Business Problem & Objective

Understanding how a person is feeling — from a photo, a video call, or a live camera — is valuable across many situations: gauging customer satisfaction during a service interaction, supporting mental health and wellbeing monitoring, or making digital products feel more natural and responsive. Today, this kind of read is either done manually (slow, inconsistent, and doesn't scale) or not done at all.

The objective of this project was to build an automated system that can reliably classify a person's facial expression into one of seven common emotional categories — angry, disgust, fear, happy, neutral, sad, and surprise — from a single image, and to prove this works in a live, real-time setting, not just on curated test data.

What success looks like

- A trained model that performs meaningfully better than guessing, and in line with what independent, published research achieves on the same standard task.
- A working application — not just a model in a notebook — that can be tried, demonstrated, and piloted.
- An honest, evidence-based understanding of where the system performs well and where it currently has limits, so any future investment decision is well-informed.

3. Approach & Methodology

We used a large, publicly recognised reference dataset of roughly 36,000 labelled face images (the standard benchmark used across the industry for this exact task), split into a training set the system learns from and a separate test set used purely to check its performance afterwards, on data it had never seen.

Rather than committing to a single approach, we built and fairly compared six different modelling strategies — ranging from a custom-built model designed specifically for this dataset, to several “pretrained” models that had already learned general visual features from millions of everyday images and were then specialized for emotion recognition. This comparison approach reduces the risk of settling on a weaker solution simply because it was the first one tried.

Process, in plain terms

- Data preparation: checked the dataset for quality issues (it was clean, with one notable imbalance — far fewer “disgust” examples than other emotions — which we corrected for during training).
- Model building: trained a custom model from scratch as a baseline, then several “transfer learning” models that build on proven, pre-existing AI vision systems.
- Fine-tuning: took the strongest pretrained model and specialized it further specifically for facial expressions.
- Fair comparison: measured every model the same way, on the same untouched test data, and selected the one with the best, most balanced performance across all seven emotions — not just the ones that are easiest to detect.
- Real-world build-out: added automatic face detection, then packaged the winning model into a live webcam tool and a web application.

4. Key Results & Metrics

The table below summarizes how each approach performed on the untouched test data. “Accuracy” is the plain-language headline number — the percentage of images the system correctly identified the emotion for.

Approach	Accuracy	Business Read
Final selected model (Fine-tuned pretrained AI)	63.2%	Best overall balance across all 7 emotions — selected for deployment
Custom-built model	63.3%	Nearly identical accuracy, but weaker on harder-to-read emotions
Tuned version of custom model	61.0%	Automated optimization did not beat the original — a useful, honest finding
Pretrained AI (before specialization)	49–56%	Confirms specialization step added real, measurable value

Translating accuracy into business terms

- Correctly reads the right emotion roughly 2 times out of 3 — far above random guessing (which would succeed only 1 time in 7, since there are seven emotion categories).
- This result is in line with independently published research benchmarks on this same standard dataset, where even human reviewers only agree with each other on the “correct” label roughly 65–70% of the time — some expressions are genuinely ambiguous, for people and AI alike.
- The system is notably strong on clear, common expressions such as happiness and surprise, and weaker on more subtle ones such as sadness and fear — a pattern consistent with human difficulty on the same images.
- Live camera testing confirmed the system works outside the lab: in good lighting with a clear expression, it reached as high as 98% confidence on a correct read; results are more variable under poor lighting or subtle expressions, discussed in Section 7.

5. Visuals

The following charts support the results above and are recommended for inclusion in any live presentation of this report.

[CHART PLACEHOLDER]

Figure 1 — Model Performance Comparison

Bar chart: accuracy (%) for each of the 6 approaches evaluated, from Section 4

[CHART PLACEHOLDER]

Figure 2 — Per-Emotion Accuracy Breakdown

Bar chart: how reliably the final model identifies each of the 7 emotions individually

[CHART PLACEHOLDER]

Figure 3 — Before / After: Real-World Optimization

Comparison chart: model confidence on live camera input before vs. after real-world tuning (Section 7)

6. Business Impact & ROI

Efficiency

- Replaces manual, one-by-one visual review of expressions with an automated read that can run continuously, at scale, and in real time.
- A live camera feed is analysed and labelled several times per second, with no ongoing manual effort once deployed.

Cost & Scale

- Built entirely on freely available reference data and open-source tools — no licensing cost was incurred to reach this stage.
- The approach is repeatable: additional use cases (e.g. a different camera setup, a different customer touchpoint) can reuse the same trained model with modest integration effort, rather than starting over.

Risk Reduction

- Because the system's accuracy and specific weak points (Section 7) are now documented with evidence rather than assumed, any decision to expand its use can be made with clear eyes on where it is reliable and where it currently is not.

Illustrative use cases this unlocks

- Aggregated, anonymized sentiment signal during customer service interactions, to flag patterns for follow-up rather than relying solely on post-hoc surveys.
- A lightweight, automated wellbeing check-in signal in supportive/monitoring contexts.
- More responsive, emotion-aware digital interfaces or kiosks.

Note: quantified cost savings and revenue impact are not yet available, as the system has not yet been piloted against a specific, measured business process. We recommend a scoped pilot (Section 8) to produce real ROI figures before a wider investment decision.

7. Challenges & Limitations

We want to be direct about where the system currently stands, so expectations are well-calibrated for any next decision.

- Accuracy ceiling: at ~63%, roughly one in three reads will be wrong on the hardest, most ambiguous expressions — consistent with independent published research on this same task, not unique to our system.
- Certain emotions are harder than others: sadness, fear, and neutral expressions are the most frequently confused with one another, both by our system and in published human-agreement studies on the same data.
- Real-world conditions add variability: lighting, camera angle, and image sharpness all affect live results more than they affect performance on the clean, curated benchmark data. We identified and fixed several real-world issues during testing, which measurably improved live performance, though some sensitivity remains (see Appendix, 9.5).
- One of the six approaches evaluated (a widely-used but computationally heavy model, VGG16) could not be fully tested due to free-tier compute limits on the platforms used — this is a resourcing constraint, not a result, and is flagged for completeness rather than hidden.
- The system currently runs on standard computer hardware (no specialized graphics acceleration), which is adequate for the current pilot-scale use but would need to be revisited for high-volume, always-on deployment.

None of the above are blockers to a scoped pilot; they are the right inputs for deciding how and where to pilot first.

8. Next Steps / Recommendations

- Run a scoped, low-risk pilot in one specific use case (e.g. a single customer touchpoint or internal tool) to gather real usage data and produce concrete ROI figures.
- Collect additional real-condition examples from the intended deployment environment (specific lighting, camera, typical distance) and use them to further specialize the model — this is the single highest-leverage next step for improving live accuracy.
- Revisit hardware requirements (standard vs. accelerated) once expected usage volume for the pilot is known.
- Formally complete evaluation of the one remaining approach (VGG16) on adequate compute, to close out the full comparison.
- Define specific success metrics for the pilot (e.g. target accuracy for the pilot's specific use case, acceptable response time) before scaling further.

9. Appendix — Technical Reference

The following technical detail is provided for reference and is not required reading for a business decision.

9.1 Dataset

- ~36,000 labelled grayscale face images, 7 emotion categories, split into training and test sets.
- One class (“disgust”) was significantly underrepresented (~1.5% of the data) and was corrected for using weighted training rather than by discarding or duplicating data.

9.2 Approaches Evaluated

- Custom convolutional neural network (CNN), built and tuned specifically for this dataset.
- Automated hyperparameter search on the custom CNN (Keras Tuner).
- Four pretrained computer-vision backbones (MobileNetV2, ResNet50, VGG16, DenseNet121), specialized via transfer learning.
- Fine-tuning of the strongest pretrained backbone (ResNet50), selected as the final model.

9.3 Final Model Performance Detail

Metric	Value
Test accuracy	63.17%
Weighted precision	0.64
Weighted recall	0.63
Weighted F1-score	0.63

9.4 Application Stack

- Face detection: OpenCV Haar Cascade.
- Model framework: TensorFlow / Keras.
- Deployment: Streamlit web application (photo upload and live webcam via WebRTC), model hosted on Hugging Face Hub.

9.5 Real-World Tuning Applied

- Face-crop padding and aspect-ratio correction, to match the framing the model was trained on.
- Lighting/contrast normalization per frame.
- Resolution matching between live camera input and training-data detail level.
- Frame-to-frame smoothing and confidence thresholds, to reduce flicker in the live video display.